***Solution:***

ABC is a triangle and G is its centroid. If,

$AD = 12\,cm \Rightarrow GD = \frac{1}{3}$ of $AD = \frac{1}{3}(12) = 4\,cm$ and $BE = 9\,cm \Rightarrow BG = \frac{2}{3}$ of $BE = \frac{2}{3}(9) = 6\,cm$.

Also D is a midpoint of BC $\Rightarrow BD = \frac{1}{2}$ of $BC = \frac{1}{2}(8) = 4\,cm$.

$\therefore$ The perimeter of $\Delta BDG = BD + GD + BG = 4 + 4 + 6 = 14$ cm

## 5.7 Altitude of a Triangle

Altitude of a triangle also known as the height of the triangle, is the perpendicular drawn from the vertex of the triangle to the opposite side. The altitude makes a right angle with the base of a triangle. Here, in $\Delta ABC$, AD is one of the altitudes as AD $\perp$ BC.

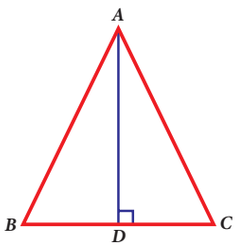
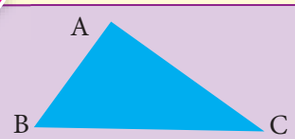
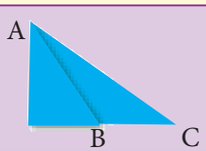
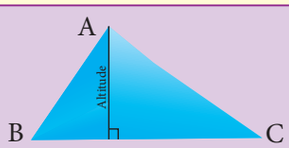


Fig. 5.34

### Activity

| | | |
|---|---|---|
|   |   |   |
| 1. Consider a paper cut-out of an acute angled triangle. Name it, say ABC. | 2. Fold the triangle so that a side overlaps itself and the fold contains the vertex opposite to that side. | 3. You can now draw the altitude AM, if you want to see it clearly. |

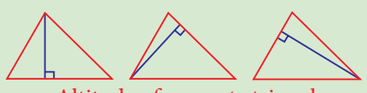
In the same way, you find altitudes of other two sides. Also, with the help of your teacher, you find altitudes of right angled triangle and obtuse angled triangle. Do the altitudes of triangle pass through the same point? What is your conclusion? We see that,

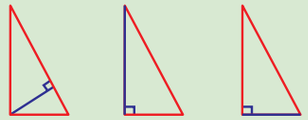
**The three altitudes of *any* triangle are concurrent.**

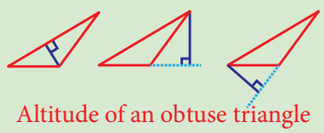
The point of concurrence is known as its Orthocentre, denoted by the letter H.

### Think

1. **In any acute angled triangle,** all three altitudes are inside the triangle. Where will be the orthocentre? In the interior of the triangle or in its exterior?
2. **In any right angled triangle,** the altitude perpendicular to the hypotenuse is inside the triangle; the other two altitudes are the legs of the triangle. Can you identify the orthocentre in this case?
3. **In any obtuse angled triangle,** the altitude connected to the obtuse vertex is inside the triangle, and the two altitudes connected to the acute vertices are outside the triangle. Can you identify the orthocentre in this case?


Altitude of an acute triangle


Altitude of a right triangle


Altitude of an obtuse triangle



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